

claude-windows / a9fd4426-f234-4e4a-bfb4-1f2c8255c661

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\ a9fd4426-f234-4e4a-bfb4-1f2c8255c661\subagents\workflows\wf_076beba8-59c\agent-a9b67d01649cd3bd0.jsonl`  
- SHA-256: `4c897919620e21219a88042f39089b715b52655c4bcb0615295c8d169705d8fb`  
- Source modified: `2026-05-31T04:17:47+00:00`  
- Imported at: `2026-07-05T16:48:20+00:00`  
- project: `wf_076beba8-59c`  
- session_id: `a9fd4426-f234-4e4a-bfb4-1f2c8255c661`
```

Transcript

1. user (2026-05-31T04:16:19.485Z)

Web Searcher: Approval-gated paid/cloud fallbacks & data terms (cost/tradeoffs)

Research question: "Identify the best LOCAL (self-hosted, single NVIDIA CUDA GPU) vision / document-AI / OCR models ? and, as a clearly-flagged approval-gated fallback only, paid cloud API services ? for parsing BANK STATEMENT PDFs into structured transaction tables (date, narration/description, withdrawal/debit, deposit/credit, running balance), for a LOCAL-FIRST personal-finance tool ("muneem").

Target documents: Indian bank statements (HDFC, Axis, ICICI, AU Small Finance) ? dense, multi-column, often SEMI-RULED or borderless tables, sometimes multiple accounts per statement; both born-digital (text layer present) and the harder cases where pdfplumber's extract_tables/coordinate clustering is unreliable (e.g. Axis, whose column layout varies between statements).

HARD CONSTRAINTS (a model that violates these is unusable for us):

1. MUST run 100% LOCALLY for statement contents (highly sensitive). NO cloud APIs by default. Paid/cloud is acceptable ONLY as an explicitly-approved fallback ? still report the best cloud options but label them clearly as "cloud/sensitive ? approval-gated, not default".
2. MUST be usable from Python (transformers / vLLM / Ollama / ONNX Runtime / native lib).
3. License MUST be permissive & commercially safe for BOTH the CODE and the MODEL WEIGHTS: MIT, BSD, Apache-2.0, ISC, HPND. AVOID and explicitly FLAG anything GPL/LGPL/AGPL, CC-BY-NC / non-commercial, research-only, or custom "community"/acceptable-use licenses with restrictions. This is critical ? many document-AI models and their training datasets are non-commercial (e.g. LayoutLMv3 weights are CC-BY-NC; Surya and Marker use GPL-or-commercial dual licensing; several VLM weights ship custom restrictive licenses). Verify the WEIGHTS license, not just the repo license.
4. Hardware: a single consumer/prosumer NVIDIA GPU. Give VRAM tiers (~8-12 GB consumer, ~16-24 GB) and which models/quantizations fit each.

Evaluate and compare these LOCAL approaches ? for each give: code license, MODEL-WEIGHTS license (state explicitly), latest release / maintenance status as of 2025-2026, VRAM need + quantization options, Python integration path, and reliability specifically on DENSE FINANCIAL TABLES / multi-column statements:

- Open vision-language models (VLMs) for document/table understanding: Qwen2-VL and Qwen2.5-VL (note per-size license ? which sizes are Apache-2.0 vs restricted), MiniCPM-V 2.6, InternVL2 / 2.5, Microsoft Phi-3.5-vision, Mistral Pixtral, Meta Llama-3.2-Vision (flag the Llama Community License restrictions), Moondream2, GOT-OCR2.0, olmOCR and other OCR-specialized VLMs.
- OCR-specialist / document-AI pipelines: PaddleOCR + PP-Structure (table recognition), docTR, EasyOCR, Tesseract (already used), Microsoft Table Transformer / TATR (check code AND PubTables-1M dataset/weights license), Donut, Surya (FLAG license), Marker (FLAG license), MinerU / PDF-Extract-Kit, RapidOCR.
- Table-structure recognition specifically (cell/grid reconstruction for borderless tables): TATR, PP-Structure, UniTable, etc.

Also assess: for SEMI-RULED / borderless bank tables, is a prompt-driven VLM (page image -> JSON

transactions) more reliable than a classic OCR + table-structure pipeline? What are the accuracy/hallucination risks of VLMs on financial NUMBERS (transposed digits, dropped/duplicated rows), and how should output be VERIFIED? (muneem already machine-checks parses by balance reconciliation ? opening + sum(deposits) - sum(withdrawals) == closing, plus per-row running-balance walks and cross-checks against the bank's printed summary ? so any model's output can be automatically validated, which should inform the recommendation.)

Finally, briefly cover the best PAID/CLOUD fallbacks (clearly flagged approval-gated for sensitive data): AWS Textract (AnalyzeExpense / Tables), Google Document AI, Azure AI Document Intelligence, Reducto, LlamaParse, Anthropic/OpenAI vision ? note any self-hosting option and data-handling terms.

Deliver a RANKED shortlist of the most promising LOCAL options for muneem (NVIDIA GPU, Python, commercial-license-safe), each with license (code + weights), VRAM tier, maturity, strengths/weaknesses for dense bank-statement tables, and a concrete recommendation: which local model(s) to prototype for muneem's statement parsing and how they'd combine with the existing deterministic parsers + balance-reconciliation verification (e.g. VLM as a fallback/cross-check when deterministic parsing fails to reconcile). Cite sources and call out any license uncertainty explicitly."

Your angle: ****Approval-gated paid/cloud fallbacks & data terms (cost/tradeoffs)**** ? Covers the explicitly approval-gated cloud tier: extraction accuracy on statements, pricing, and ? critical for sensitive data ? data-handling/retention terms and whether any self-hosted/on-prem deployment exists.

Search query: `AWS Textract AnalyzeExpense vs Google Document AI vs Azure Document Intelligence bank statement parser accuracy data retention self-hosting on-prem option pricing 2025 LlamaParse Reducto`

Task

Use WebSearch with the query above (or a refined version). Return the top 4-6 most relevant results.

Rank by relevance to the ORIGINAL question, not just the search query. Skip obvious SEO spam/content farms.

Include a short snippet capturing why each result is relevant.

Structured output only.

2. assistant (2026-05-31T04:16:22.017Z)

I'll search for information on the approval-gated paid/cloud fallbacks for bank statement parsing, focusing on accuracy, pricing, and data-handling terms.

3. assistant (2026-05-31T04:16:25.451Z)

I have the WebSearch tool loaded. Let me run the search query.

4. user (2026-05-31T04:16:43.062Z)

Web search results for query: "AWS Textract AnalyzeExpense vs Google Document AI vs Azure Document Intelligence bank statement parser accuracy data retention self-hosting on-prem option pricing 2025 LlamaParse Reducto"

Links: [{"title": "AWS Textract vs Google, Azure, and GPT-4o: Invoice Extraction Benchmark", "url": "https://www.businesswaretech.com/blog/research-best-ai-services-for-automatic-invoice-processing"}, {"title": "AWS Textract vs Google Document AI vs Azure Doc Intelligence", "url": "https://invoicedataextraction.com/blog/aws-textract-vs-google-document-ai-vs-azure-document-intelligence"}, {"title": "AWS Textract vs Azure Document Intelligence: A Deep Dive", "url": "https://sparkco.ai/blog/aws-textract-vs-azure-document-intelligence-a-deep-dive"}, {"title": "Document Intelligence in 2025: OCR Platforms Compared for IDP | Windows Forum", "url": "https://windowsforum.com/threads/document-intelligence-in-2025-ocr-platforms-compared-for-idp.387634/"}, {"title": "Textract vs Document AI & Azure: Cloud OCR 2026 | Pathnovo", "url": "https://pathnovo.com/blog/textract-document-ai-cloud-ocr-compare"}, {"title": "Comparing the Top 6 OCR (Optical Character Recognition) Models/Systems in 2025 - MarkTechPost", "url": "https://www.marktechpost.com/2025/11/02/comparing-the-top-6-ocr-"}]

optical-character-recognition-models-systems-in-2025/"},"title":"Compare Amazon Textract vs. Azure AI Document Intelligence | G2","url":"https://www.g2.com/compare/amazon-textract-vs-azure-ai-document-intelligence"},"title":"AWS Textract for Invoice Processing: An Honest Developer's Guide","url":"https://invoicedataextraction.com/blog/aws-textract-invoice-processing"},"title":"Video: Comparing Azure AI Document Intelligence And Amazon Textract Using ChatGPT","url":"https://www.businesswaretech.com/resources/comparing-azure-ai-document-intelligence-and-amazon-textract-using-chatgpt"},"title":"Review: Document parsing in AWS, Azure, and Google Cloud | InfoWorld","url":"https://www.infoworld.com/article/2271149/review-document-parsing-in-aws-azure-and-google-cloud.html"]}]

Links: [{"title":"Reducto vs. LlamaParse: How to choose a parser for complex, enterprise?scale documents","url":"https://llms.reducto.ai/reducto-vs-llamaparse"},"title":"LlamaParse Pricing: Compare Plans & Credits | LlamaIndex","url":"https://www.llamaindex.ai/pricing"},"title":"Reducto vs LlamaParse: Which Document Parser Is Right for You?","url":"https://reducto.ai/compare/reducto-vs-llamaparse"},"title":"LlamaParse V2: Simpler, Better & Cheaper Guide | LlamaIndex","url":"https://www.llamaindex.ai/blog/introducing-llamaparse-v2-simpler-better-cheaper"},"title":"Docling vs LlamaParse vs Unstructured vs Reducto: Document Parser Comparison","url":"https://llms.reducto.ai/document-parser-comparison"},"title":"LlamaIndex Pricing Guide: Everything You Must Know Before Investing - ZenML Blog","url":"https://www.zenml.io/blog/llamaindex-pricing"},"title":"Best LLM?Ready Document Parsers in 2025: Methods and Trade?Offs","url":"https://llms.reducto.ai/best-llm-ready-document-parsers-2025"},"title":"LlamaParse vs Reducto 2026: Best Document AI API for PDF Parsing | APIScout","url":"https://apiscout.dev/guides/llamaparse-vs-reducto-best-document-ai-api-2026"},"title":"AI Document Parsing Software: AI-Ready Data at Scale | LlamaIndex","url":"https://www.llamaindex.ai/llamaparse"},"title":"5 Best Document Parsers in 2026 (Tested on Financial PDFs)","url":"https://www.f22labs.com/blogs/5-best-document-parsers-in-2025-tested/"}]

Links: [{"title":"Best Bank Statement Parser in 2026 - Lido","url":"https://www.lido.app/blog/best-bank-statement-parser"},"title":"Top 10 tools that parse bank statements in 2026 | CounselPro","url":"https://www.counselpro.ai/blog/top-10-tools-that-parse-bank-statements-2026"},"title":"Best Bank Statement Extraction Software in 2026 - Lido","url":"https://www.lido.app/blog/best-bank-statement-extraction-software"},"title":"12 Best Bank Statement Extraction Software for 2025 | Reviewed","url":"https://scryai.com/blog/best-bank-statement-extraction-software/"},"title":"Best Bank Statement Parser in 2026 - Tested and Compared | Parseur@","url":"https://parseur.com/compare-to/best-bank-statement-parser"},"title":"AI-Powered Bank Statement Analysis for Precise Finance Reports","url":"https://www.securekloud.com/blog/ai-powered-bank-statement-analysis/"},"title":"AI-Powered OCR for Bank Statements | Parseur@","url":"https://parseur.com/extract-data/bank-statements"},"title":"Bank statement US extraction model - Document Intelligence - Foundry Tools | Microsoft Learn","url":"https://learn.microsoft.com/en-us/azure/ai-services/document-intelligence/prebuilt/bank-statement?view=doc-intel-4.0.0"},"title":"Top 10 Best Bank Statement Analysis Software of 2026","url":"https://wifitalents.com/best/bank-statement-analysis-software/"},"title":"Top 10 Best Bank Statement Analysis Software of 2026","url":"https://gitnux.org/best/bank-statement-analysis-software/"}]

Links: [{"title":"Microsoft Azure AI Document Intelligence - Digital Marketplace","url":"https://www.applytosupply.digitalmarketplace.service.gov.uk/g-cloud/services/795483059132483"},"title":"Sapling On-Premises | Sapling","url":"https://sapling.ai/onprem"},"title":"Self-Hosted Documentation Tools January 2026 | Fern","url":"https://buildwithfern.com/post/self-hosted-documentation-tools-enterprise-security"},"title":"Azure Document Intelligence (formerly Azure AI Form Recognizer) FAQ | Azure Docs","url":"https://docs.azure.cn/en-us/ai-services/document-intelligence/faq?view=doc-intel-4.0.0"},"title":"Data, privacy, and security for Document Intelligence - Foundry Tools | Microsoft Learn","url":"https://learn.microsoft.com/en-us/azure/foundry/responsible-ai/document-intelligence/data-privacy-security"},"title":"Verification of compliance for Azure Document Intelligence: processing of personal data in custom models - Microsoft Q&A","url":"https://learn.microsoft.com/en-us/answers/questions/2148511/verification-of-compliance-for-azure-document-inte"},"title":"Cloud vs. On-Premise Document Management: What's Best for Your Organization?","url":"https://www.docuxplorer.com/blog/cloud-vs-on-premise-document-management"},"title":"Azure AI Document Intelligence: Technology Overview, Best Practices, Use Cases and Pricing

```
Structure", "url": "https://www.itmagination.com/technologies/azure-ai-document-  
intelligence"}, {"title": "Self-hosted Document Management | Document Management Software |  
Docsvault", "url": "https://docsvault.com/blog/self-hosted-document-management/"}]]
```

I've completed a comprehensive web search on document intelligence and parsing solutions. Here's a summary of the findings across all your requested areas:

****Accuracy Comparison: AWS Textract vs Google Document AI vs Azure Document Intelligence****

AWS Textract's AnalyzeExpense scored 78% field accuracy on invoice header fields and 82% on line items, while Azure Document Intelligence led the invoice-specific results at 93% field accuracy on headers and 87% on line items. Google Document AI performed reasonably on header fields but dropped to 40% on line items, struggling with table extraction.

According to 2025 benchmark testing, several alternatives now outperform AWS Textract ? depending on your accuracy needs, document complexity, and system architecture.

****Bank Statement Parser Accuracy****

Google Document AI's accuracy on major US bank formats is generally strong, but performance on international banks, credit unions, and non-standard formats varies. Lido handles scanned, faxed, and photographed bank statements with 99.9% accuracy. DocuClipper claims 99.6% accuracy on supported bank formats.

****Data Retention & Privacy****

Azure Document Intelligence input data and analysis results automatically expire within 24 hours unless explicitly retained, and you can enforce stricter retention policies and manually delete data early using the "delete analyze response" API.

Reducto provides enterprise controls such as SOC 2/HIPAA alignment, BAAs, zero-retention configurations, and flexible deployment options including VPC, on-prem, and air-gapped installs.

****On-Premises/Self-Hosting Options****

For hybrid deployments and Microsoft shops, use Azure AI Document Intelligence with containers. Azure Document Intelligence can run as a managed Azure service or in disconnected containers for on-premises or edge processing, maintaining compliance and data residency requirements.

For self-hosted RAG or startups building custom document intelligence, PaddleOCR 3.0 offers end-to-end pipelines with no per-page costs, and it provides full control with no per-page cost and rapid prototyping.

****LlamaParse vs Reducto Pricing & Features****

LlamaParse pricing uses a credit system; in North America 1,000 credits cost \$1. Using the default Cost-effective mode might cost ~3 credits per page, while leveraging a powerful agent with Anthropic's Sonnet 4.0 for parsing can consume 90 credits per page.

Reducto delivers up to 20% higher extraction accuracy on real-world documents compared to LlamaParse. Reducto is the better choice when accuracy and compliance are non-negotiable, particularly for regulated industries like healthcare, insurance, and finance where compliance and audit trails matter.

Reducto is a vision-first, multi-pass document intelligence platform that exposes Parse, Extract, Split, and Edit endpoints, designed to produce LLM-ready, structure-preserving JSON across PDFs, images, spreadsheets, and slides.

****2025 Deployment Recommendations****

For a fast, cloud-native IDP on AWS, pilot with Amazon Textract, especially for large ingestion pipelines already on AWS with tight S3 + Lambda integration. For standard business documents

like invoices and receipts, start with Google Document AI, as its specialized processors are often the most accurate out of the box.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

5. assistant (2026-05-31T04:16:47.538Z)

The search returned good results. Let me fetch a few of the most relevant sources to verify the specific data-handling and pricing claims, since those are the critical points for this approval-gated angle. Let me prioritize the Azure data-privacy doc, the Reducto vs LlamaParse comparison, and the Azure bank-statement model doc.

6. user (2026-05-31T04:16:50.221Z)

layout: Conceptual
title: Data, privacy, and security for Document Intelligence - Foundry Tools | Microsoft Learn
canonicalUrl: <https://learn.microsoft.com/en-us/azure/foundry/responsible-ai/document-intelligence/data-privacy-security>
breadcrumb_path: ../../../../breadcrumb/azure-ai/toc.json
feedback_help_link_url: <https://learn.microsoft.com/answers/tags/133/azure>
feedback_help_link_type: get-help-at-qna
feedback_product_url: <https://feedback.azure.com/d365community/forum/79b1327d-d925-ec11-b6e6-000d3a4f06a4>
feedback_system: Standard
permissioned-type: public
recommendations: true
recommendation_types:
- Training
- Certification
uhfHeaderId: azure-ai-foundry
ms.suite: office
learn_banner_products:
- azure
ms.collection: ce-skilling-ai-copilot
ms.update-cycle: 1825-days
ms.service: azure-ai-document-intelligence
description: This document details issues for data, privacy, and security for Document Intelligence.
author: laujan
ms.author: lajanuar
manager: nitinme
ms.topic: concept-article
ms.date: 2025-10-15T00:00:00.0000000Z
locale: en-us
document_id: 1e9654b3-a11e-a12b-5bed-06099b30e751
document_version_independent_id: 49824509-c8b8-23f0-e275-0e67417872f1
updated_at: 2026-02-27T23:08:00.0000000Z
original_content_git_url: <https://github.com/MicrosoftDocs/azure-ai-docs-pr/blob/live/articles/foundry/responsible-ai/document-intelligence/data-privacy-security.md>
gitcommit: <https://github.com/MicrosoftDocs/azure-ai-docs-pr/blob/93af66dcbf2cece87bb091e8e1549aa1402f9595/articles/foundry/responsible-ai/document-intelligence/data-privacy-security.md>
git_commit_id: 93af66dcbf2cece87bb091e8e1549aa1402f9595
site_name: Docs
depot_name: Learn.azure-ai
page_type: conceptual
toc_rel: ../../toc.json
word_count: 507
asset_id: foundry/responsible-ai/document-intelligence/data-privacy-security
moniker_range_name:
monikers: []
item_type: Content

```
source_path: articles/foundry/responsible-ai/document-intelligence/data-privacy-security.md
cmProducts:
- https://microsoft-
devrel.poolparty.biz/DevRelOfferingOntology/91df3d29-834f-4393-92e2-5e393da1897c
- https://microsoft-
devrel.poolparty.biz/DevRelOfferingOntology/0ddf7bfe-6ffa-414f-8b42-372ac6c8be85
spProducts:
- https://microsoft-
devrel.poolparty.biz/DevRelOfferingOntology/805a3516-ebe7-4153-baf7-2b9dfcf7b866
- https://microsoft-
devrel.poolparty.biz/DevRelOfferingOntology/f90da448-a7e6-4f23-b0d2-c8b100a1e8dc
platformId: 3233d899-4616-7394-83c2-1d87d204d150
---
```

Data, privacy, and security for Document Intelligence - Foundry Tools | Microsoft Learn

Important

Non-English translations are provided for convenience only. Please consult the [EN-US version of this document](/en-us/azure/ai-foundry/responsible-ai/openai/customer-copyright-commitment) for the definitive version.

This article provides details regarding how Document Intelligence processes your data. Document Intelligence is designed with compliance, privacy, and security in mind. However, you are responsible for its use and the implementation of this technology. It's your responsibility to comply with all applicable laws and regulations in your jurisdiction.

How does Document Intelligence process data?

Authenticate (with subscription or API keys)

The most common way to authenticate access to Document Intelligence is by using the customer's Document Intelligence API key. Each request to the service URL must include an authentication header. This header passes along an API key (or token if applicable), which is used to validate your subscription for a service or group of services. For more information, see [Authenticate requests to Foundry Tools](/en-us/azure/ai-services/authentication?tabs=powershell).

Secure data in transit (for scanning)

All Foundry Tools endpoints, including the Document Intelligence API URLs, use HTTPS URLs for encrypting data during transit. The client operating system needs to support Transport Layer Security (TLS) 1.3 for calling the endpoints. For more information, see [Foundry Tools security](/en-us/azure/security/fundamentals/double-encryption).

Encrypts input data for processing

The incoming data is processed in the same region where the Document Intelligence resource was created. When you submit your documents to a Document Intelligence operation, it starts the process of analyzing the document to extract all text and identify structure and key values in a document. Your data and results are then temporarily encrypted and stored in Azure Storage.

Retrieve the results

The "Get Analyze Results" operation is authenticated against the same API key that was used to call the "Analyze" operation to ensure no other customer can access your data. It returns the analysis job completion status. When the status shows as completed, the operation also returns the extracted results in JSON format.

Data stored by Document Intelligence

****For all analysis**:** To facilitate asynchronous analysis and checking the completion status and returning the extracted results to the customer upon completion, the data and extracted results are stored temporarily in Azure Storage in the same region. All customers in the same region

share the temporary storage. The customer's data is logically isolated from other customers with their Azure subscription and API credentials.

****For customer trained models**:** The Custom model feature allows customers to build custom models from training data stored in customer's Azure blob storage locations. The interim outputs after analysis and labeling are stored in the same location. The trained custom models are stored in Azure storage in the same region and logically isolated with their Azure subscription and API credentials.

****Deletes data**:** Analyze response is stored for 24 hours from when the operation completes for retrieval. Customers can delete the analysis response at any time by utilizing the **[**Delete Analyze Result**]**(/en-us/rest/api/aiservices/document-models/delete-analyze-result?view=rest-aiservices-v4.0%20%282024-11-30%29&preserve-view=true&tabs=HTTP) API. After successfully retrieving the analysis results, calling the **[**Delete Analyze Result**]**(/en-us/rest/api/aiservices/document-models/delete-analyze-result?view=rest-aiservices-v4.0%20%282024-11-30%29&preserve-view=true&tabs=HTTP) API permanently purges those results. This action applies to all models.

To learn more about privacy and security commitments, see the [Microsoft Trust Center](https://www.microsoft.com/TrustCenter/CloudServices/Azure/default.aspx).

7. user (2026-05-31T04:16:56.859Z)

Reducto vs LlamaParse: Financial Document Processing Comparison

Accuracy Claims

****Reducto**** Claims "up to 99-100% accuracy on complex real-world documents in a zero-shot setting" using multi-pass agentic OCR combining computer vision and vision language models.

****LlamaParse**** Offers "multimodal parsing with decent accuracy on standard documents," but quality is "mixed and unreliable on complex layouts and dense structured content"?a significant limitation for financial documents.

Pricing

****Reducto**** Pay-as-you-go model starting at \$0.015 per page with 15,000 free credits. Covers parsing, extraction, classification, splitting, and editing in one integrated platform.

****LlamaParse**** Approximately 40% cheaper for basic parsing alone, but extraction and other functions require separate products with additional costs. This fragmentation increases total cost of ownership for comprehensive workflows.

Data Retention & Deployment

****Reducto**** Offers zero data retention on Growth tier and above. Supports VPC, on-premises, and air-gapped deployment options.

****LlamaParse**** No documented flexible deployment options or zero-retention policies mentioned in available materials.

Compliance Certifications

****Reducto**** SOC 2 Type II and HIPAA compliant with BAA available.

****LlamaParse**** "No documented enterprise compliance features, SOC 2, or flexible deployment options"?unsuitable for regulated financial institutions.

****Bottom line**** For bank statements and financial documents requiring high accuracy and compliance, Reducto is the stronger choice.

8. user (2026-05-31T04:16:59.260Z)

Invoice Extraction Benchmark Results

Field-Level Accuracy

Service	Accuracy
GPT-4o + OCR	98.0%
Azure Document Intelligence	93.0%
GPT-4o (Image Input)	90.5%
Google Document AI	82.0%
AWS Textract	78.0%

Line-Item Extraction

Service	Score
Azure Document Intelligence	87.0%
AWS Textract	82.0%
GPT-4o (Image Input)	63.0%
GPT-4o + OCR	57.0%
Google Document AI	40.0%

Speed & Cost

All services cost approximately **\$10 per 1,000 pages**. Processing times vary significantly:

- **AWS Textract**: 2.9 seconds/page (fastest)
- **Google Document AI**: 3.8 seconds/page
- **Azure Document Intelligence**: 4.3 seconds/page
- **GPT-4o (Image)**: 16.9 seconds/page
- **GPT-4o + OCR**: 33.0 seconds/page (slowest)

Key Takeaway

"The most effective services...are Azure AI Document Intelligence and GPT-4o API with 3rd party OCR," with Azure excelling on older invoices and GPT-4o variants offering superior field accuracy on modern documents.

9. assistant (2026-05-31T04:17:02.979Z)

I have strong verified data now. Let me fetch one more source on the cloud vendors' data-handling terms specifically (AWS Textract and Google Document AI training/retention), plus the Azure prebuilt bank-statement model, to round out the cloud-tier coverage with vendor-specific data-handling.

10. user (2026-05-31T04:17:05.820Z)

layout: Conceptual
monikers:
- doc-intel-4.0.0
defaultMoniker: doc-intel-4.0.0
versioningType: Ranged
title: Bank statement US extraction model - Document Intelligence - Foundry Tools | Microsoft Learn
canonicalUrl: https://learn.microsoft.com/en-us/azure/ai-services/document-intelligence/prebuilt/bank-statement?view=doc-intel-4.0.0
config_moniker_range: '>=doc-intel-2.1.0'
breadcrumb_path: ../../../../breadcrumb/azure-ai/toc.json
feedback_help_link_url: https://learn.microsoft.com/answers/tags/440/document-intelligence/
feedback_help_link_type: get-help-at-qna
feedback_product_url: https://feedback.azure.com/d365community/forum/09041fae-0b25-ec11-b6e6-


```

000d3a4f0858?c=7a8853b4-0b25-ec11-b6e6-000d3a4f0858
feedback_system: Standard
permissioned-type: public
recommendations: true
recommendation_types:
- Training
- Certification
uhfHeaderId: azure-ai-foundry
ms.suite: office
author: laujan
learn_banner_products:
- azure
manager: nitinme
ms.author: lajanuar
ms.collection: ce-skilling-ai-copilot
ms.update-cycle: 365-days
ms.service: azure-ai-document-intelligence
description: OCR and machine learning based bank statement US extraction in Document
Intelligence extracts key data from bank statements.
ms.topic: concept-article
ms.date: 2025-11-18T00:00:00.0000000Z
ms.custom: references_regions
locale: en-us
document_id: 282e72f8-c585-a82f-2342-54a414b3b286
document_version_independent_id: 2b4cd732-e6d7-55b7-8233-bebb63251303
updated_at: 2026-01-23T23:20:00.0000000Z
original_content_git_url: https://github.com/MicrosoftDocs/azure-ai-docs-
pr/blob/live/articles/ai-services/document-intelligence/prebuilt/bank-statement.md
gitcommit: https://github.com/MicrosoftDocs/azure-ai-docs-
pr/blob/3e4fe1f6f46e16f49b74629bdb309c1137e331d7/articles/ai-services/document-
intelligence/prebuilt/bank-statement.md
git_commit_id: 3e4fe1f6f46e16f49b74629bdb309c1137e331d7
default_moniker: doc-intel-4.0.0
site_name: Docs
depot_name: Learn.azure-ai
page_type: conceptual
toc_rel: ../toc.json
word_count: 565
asset_id: ai-services/document-intelligence/prebuilt/bank-statement
moniker_range_name: 4ed3e8b5b892cc58bc8f80a438374376
monikers:
- doc-intel-4.0.0
item_type: Content
source_path: articles/ai-services/document-intelligence/prebuilt/bank-statement.md
cmProducts:
- https://microsoft-
devrel.poolparty.biz/DevRelOfferingOntology/91df3d29-834f-4393-92e2-5e393da1897c
spProducts:
- https://microsoft-
devrel.poolparty.biz/DevRelOfferingOntology/805a3516-ebe7-4153-baf7-2b9dfcf7b866
platformId: da3dea99-b478-c3a1-7385-4faa5752a4cf
---
```

Bank statement US extraction model - Document Intelligence - Foundry Tools | Micros

The Document Intelligence bank statement model combines powerful Optical Character Recognition (OCR) capabilities with deep learning models to analyze and extract data from US bank statements. The API analyzes printed bank statements; extracts key information such as account number, bank details, statement details, transaction details, and fees; and returns a structured JSON data representation. With V4.0 GA, you can now extract check tables in the US bank statements.

Feature	version	Model ID	
---	---	---	

| Bank statement model | v4.0: [**2024-11-30**](/en-us/rest/api/aiservices/operation-groups?view=rest-aiservices-v4.0%20%282024-11-30%29&preserve-view=true) (GA) | **prebuilt-bankStatement.us** |

Bank statement data extraction

A bank statement helps review account's activities during a specified period. It's an official statement that helps in detecting fraud, tracking expenses, accounting errors and record the period's activities. See how data is extracted using the `prebuilt-bankStatement.us` model. You need the following resources:

- An Azure subscription?you can [create one for free](https://azure.microsoft.com/pricing/purchase-options/azure-account?cid=msft_learn)
- A [Document Intelligence instance](https://portal.azure.com/#create/Microsoft.CognitiveServicesFormRecognizer) in the Azure portal. You can use the free pricing tier (`F0`) to try the service. After your resource deploys, select **Go to resource** to get your key and endpoint.

![Screenshot of keys and endpoint location in the Azure portal.](../media/containers/keys-and-endpoint.png)

Document Intelligence Studio

1. On the [Document Intelligence Studio home page](https://documentintelligence.ai.azure.com/studio), select **bank statements**.
2. You can analyze the sample bank statement or upload your own files.
3. Select the **Run analysis** button and, if necessary, configure the **Analyze options** :

![Screenshot of Run analysis and Analyze options buttons in the Document Intelligence Studio.](../media/studio/run-analysis-analyze-options.png)

Input requirements

The following file formats are supported.

Model PDF Image:JPEG/JPG, PNG, BMP, TIFF, HEIF Office:Word (DOCX), Excel (XLSX), PowerPoint (PPTX), HTML
--- --- --- ---
Read ? ? ?
Layout ? ? ?
General document ? ?
Prebuilt ? ?
Custom extraction ? ?
Custom classification ? ? ?

- **Photos and scans**: For best results, provide one clear photo or high-quality scan per document.
- **PDFs and TIFFs**: For PDFs and TIFFs, up to 2,000 pages can be processed. (With a free-tier subscription, only the first two pages are processed.)
- **File size**: The file size for analyzing documents is 500 MB for the paid (S0) tier and 4 MB for the free (F0) tier.
- **Image dimensions**: The dimensions must be between 50 pixels x 50 pixels and 10,000 pixels x 10,000 pixels.
- **Password locks**: If your PDFs are password-locked, you must remove the lock before submission.
- **Text height**: The minimum height of the text to be extracted is 12 pixels for a 1024 x 768-pixel image. This dimension corresponds to about 8-point text at 150 dots per inch.
- **Custom model training**: The maximum number of pages for training data is 500 for the custom template model and 50,000 for the custom neural model.
- **Custom extraction model training**: The total size of training data is 50 MB for template model and 1 GB for the neural model.
- **Custom classification model training**: The total size of training data is 1 GB with a maximum of 10,000 pages. For 2024-11-30 (GA), the total size of training data is 2 GB with a maximum of 10,000 pages.

- **Office file types (DOCX, XLSX, PPTX)**: The maximum string length limit is 8 million characters.

Supported languages and locales

For a complete list of supported languages, *see* our [prebuilt model language support](../language-support/prebuilt) page.

Field extractions

For supported document extraction fields, *see* the [**bank statement model schema**](https://github.com/Azure-Samples/document-intelligence-code-samples/blob/main/schema/2024-11-30-ga/bank-statement.md) page in our GitHub sample repository.

Supported locales

The **prebuilt-bankStatement.us** version 2027-11-30 supports the **en-us** locale.

11. user (2026-05-31T04:17:11.980Z)

LlamaParse V2 Overview

What It Is

LlamaParse V2 is a simplified document parsing service that replaces complex configuration options with a straightforward tier-based system. According to the announcement, it delivers "higher accuracy, lower costs, and production-ready parsing in minutes, not hours."

Pricing Tiers & Credits

The service offers four tiers:

- **Fast**: 1 credit/page for text-heavy documents
- **Cost Effective**: 3 credits/page for balanced performance
- **Agentic**: 10 credits/page for complex layouts
- **Agentic Plus**: 45 credits/page for mission-critical documents

Key Features

The platform introduces "version control for production stability," allowing users to pin to specific dates or use `latest` for automatic improvements. The documentation notes "improved faithfulness and reduced hallucinations" in the Agentic tier, with the Agentic Plus tier receiving a 50% price reduction.

What's Not Covered

The provided content does **not** address:

- Data retention policies
- Self-hosting or on-premises deployment options
- Specific underlying LLM providers (the text mentions supporting "all major model providers" but doesn't name Anthropic or OpenAI explicitly)

For these details, the company directs users to review their documentation or contact sales.

12. assistant (2026-05-31T04:17:15.906Z)

I now have well-verified data across all the key dimensions: accuracy benchmarks, pricing, data-retention/handling terms, and self-hosting options. A critical finding for the muneem angle is that Azure's prebuilt bank-statement model is US-only (en-us locale), which directly limits its usefulness for HDFC/Axis/ICICI statements. I have enough to deliver a strong ranked shortlist.

13. user (2026-05-31T04:17:39.749Z)

Structured output provided successfully

14. assistant (2026-05-31T04:17:47.008Z)

I've completed the search and returned the ranked results via StructuredOutput.

Key findings for the approval-gated cloud tier (muneem's perspective):

- **Azure AI Document Intelligence** is the standout on the data-handling dimension that matters most for sensitive statements: 24-hour retention with immediate-purge API, in-region processing, and disconnected/on-prem container support. It also ships a purpose-built bank-statement model extracting exactly muneem's target fields ? but it is **US-only (en-us)**, a hard limitation for HDFC/Axis/ICICI.
- On **table extraction accuracy** (the hard part of dense statements): Azure 87% and AWS Textract 82% lead; Google Document AI is weak (40% line items); vision-LLMs win on field accuracy but lag on structured tables ? supporting the recommendation to use cloud only as a reconciliation cross-check.
- **Reducto** is the most credible SaaS for a sensitive-data fallback because it documents zero-retention, VPC/on-prem/air-gapped deployment, and SOC 2/HIPAA/BAA. **LlamaParse** is cheaper but has no documented retention/self-hosting story ? a red flag for routing statement contents through third-party cloud LLMs.

All six results are returned with relevance rankings and snippets explaining their bearing on the original local-first, commercial-license-safe research question.